Inverse Association Between the Skin and Oral Microbiota in Atopic Dermatitis

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Previous studies have shown independently that the skin and gut microbiota are closely associated with atopic dermatitis (AD); however, the microbiota across different habitats of AD patients as an integrated community has not been characterized. In the present study, we comparatively analyzed the structure and function of the microbial communities in the skin, oral cavity, and gut of 172 AD patients and 120 healthy controls through 16S ribosomal RNA gene amplicon sequencing. The skin and oral cavity, but not the gut, of AD patients demonstrated differential reduction in the microbial diversity, and these were distinctly correlated with disease severity. Different degrees of shifts in the community structure were found among different habitats, and the lineage distance between the skin and oral microbiota of AD patients was closer than that observed in the controls. The different habitats of AD patients exhibited site-specific alterations at the genus level, and many oral-specific microbes of AD showed opposing directions of enrichment in the skin and oral cavity. Most interestingly, an inverse association in the functional pathways was found between the skin and oral microbiota of AD patients. Additionally, the alterations of the microbiota in different body sites of AD patients were differentially affected by age.

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INTRODUCTION

Previous studies have independently shown that both the gut and skin microbiota are closely associated with AD. The gut microbiota of infant AD patients exhibits a reduced diversity (Abrahamsson et al., 2012), which is inversely related to the severity of AD (Nylund et al., 2015; Watanabe et al., 2003). The diversity of the skin microbiota in AD lesions is also significantly reduced compared with that of healthy individuals (Chng et al., 2016; Kennedy et al., 2017; Kong et al., 2012; Shi et al., 2016), and the proportion of *Staphylococcus*, particularly *Staphylococcus aureus*, is increased in AD and correlated with the severity of the disease (Byrd et al., 2017; Kong et al., 2012). However, the microbiota in different habitats of AD patients as an integrated ecological community has not been characterized, and the role of the microbiota residing in other habitats, for example, the oral cavity, of AD patients is unclear.

Different body habitats have site-specific microbiota, and these microbiota might be differentially involved in the homeostasis of an individual and in the pathogenesis of various diseases (Human Microbiome Project Consortium, 2012; Lloyd-Price et al., 2017). Under certain disease conditions, the microbial communities in multiple body sites might exhibit alterations simutaneously. For example, dysbiosis in both the gut and oral cavity has been observed in inflammatory bowel disease (Lucas et al., 2017), rheumatoid arthritis (Yeoh et al., 2013; Zhang et al., 2015), and colorectal cancer (Flynn et al., 2016), whereas asthma has been associated with alterations of the microbiota in both the gut and the respiratory tract (Huang et al., 2017). The microbiota in different habitats might contribute synergistically or oppositely to the pathogenesis of a disease, but the exact mechanism has not yet been elucidated. Thus, analyzing the microbiota in a single body habitat might be insufficient to reveal its relationship with the host.

The oral cavity in human is the second largest bacterial reservoir after the lower gastrointestinal tract (Arweiler and Netuschil, 2016) and is profoundly relevant to host health and disease (Curtis et al., 2011; Dewhirst et al., 2010). The oral microbiota is involved in the etiology of dental caries and periodontal disease (He et al., 2015), and it is also associated with chronic inflammatory diseases at distant sites, for example, inflammatory bowel disease, rheumatoid arthritis, bacterial endocarditis, atherosclerosis, respiratory diseases, and diabetes (He et al., 2015; Yeoh et al., 2013). Most recently, causal relationships between the oral microbiota and colorectal or pancreatic cancer have been confirmed (Fan et al., 2018; Flemer et al., 2018), and ectopic colonization of the oral bacteria in the intestine has been found to drive T-helper type 1 cell induction and inflammation (Atarashi et al., 2017). In addition, it has been speculated

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Abbreviations: AD, atopic dermatitis; SCORAD, Scoring of Atopic Dermatitis

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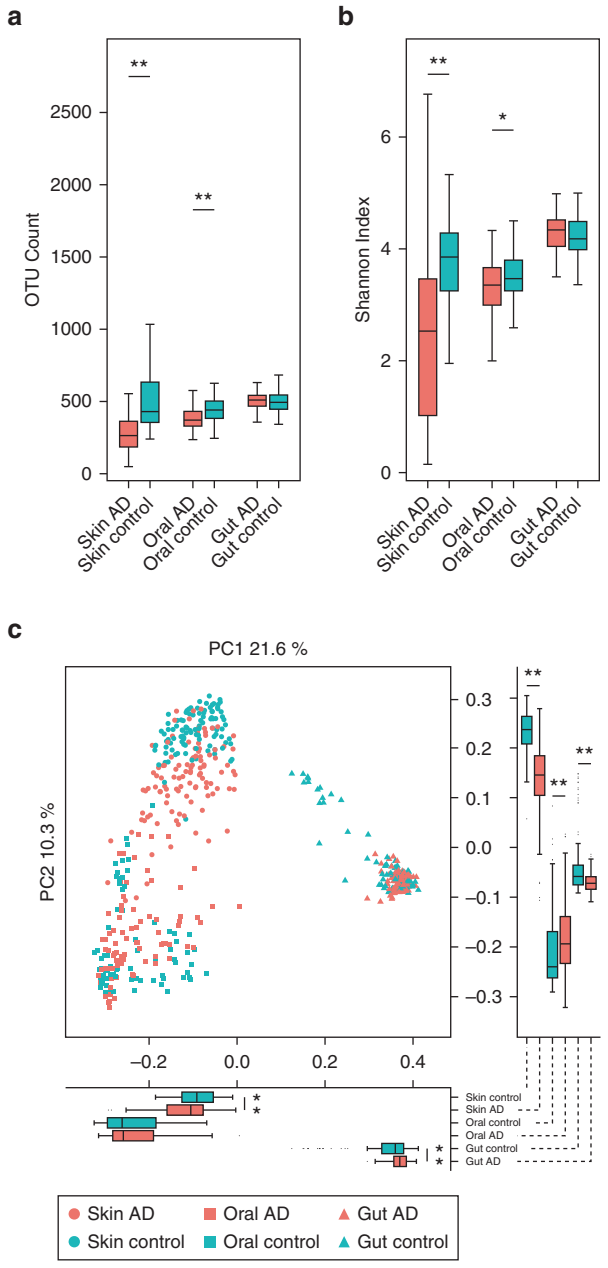


Figure 1. Community-level analysis of the microbiota in the skin, oral cavity, and gut. Richness (**a**) and α-diversity (Shannon Index), (**b**) of the microbiota in the three body sites at the OTU level. Skin (AD, n = 111, control, n = 102); oral (AD, n = 83; control, n = 77); and gut (AD, n = 88; control n = 83). *$P < 0.05$, **$P < 0.01$. A two-tailed Wilcoxon rank-sum test was used to determine significance. Boxplots represent the medians and interquartile ranges with whiskers, as determined by Tukey's method; black dots represent data points beyond the whiskers. (**c**) A principal coordinates analyses based on unweighted UniFrac distances is shown along the first two principal coordinate (PC) axes. Each point represents a single sample. Box plots indicate the distribution of samples along the given axis. AD, atopic dermatitis; OTU, operational taxonomic unit.

that oral inflammation is associated with allergic disease (Arbes and Matsui, 2011), but the characteristics of the oral microbiota in AD remain unclear.

In the present study, we aimed to comparatively analyze the microbiota in the skin, oral cavity, and gut of AD patients as an integrated community to obtain a more comprehensive understanding of the relationships among the microbiota in different habitats and the host in AD, paying additional attention to the characteristics of the oral microbiota.

RESULTS

Differentially reduced diversity in the skin and oral microbiota in AD patients

We collected 226 skin lesion samples (AD, n = 121; control, n = 105), 161 oral samples (AD, n = 84; control, n = 77), and 171 stool samples (AD, n = 88; control, n = 83) from 172 AD patients and 120 non-AD controls matched in age, sex, and body mass index (see Supplementary Table S1 online), and analyzed the 16S ribosomal RNA of bacteria from the samples. Both the operational taxonomic units richness and α-diversity of the microbiota from the skin of AD patients were significantly reduced compared with those of healthy controls ($P < 0.01$, Wilcoxon rank-sum test, Figure 1a, 1b). Specifically, the decrease in the Shannon diversity of AD patients remained significant even after the sequence of *S. aureus* was subtracted ($P < 0.01$,Wilcoxon rank-sum test, see Supplementary Figure S1 online). The operational taxonomic units richness and α-diversity of the microbiota of the oral cavity also showed significant reductions in AD patients ($P < 0.05$, Wilcoxon rank-sum test, Figure 1a, 1b). However, no difference in the operational taxonomic units richness or α-diversity of the gut microbiota was found between AD patients and healthy controls (Figure 1a, 1b). We then stratified the samples into three age groups: toddlers (1−3 years), children (4−12 years), and teenagers-adults (13−59 years), and obtained similar results for both the skin and gut samples in all age groups; whereas a reduction in α-diversity was observed in the oral cavity of the teenager-adults group (see Supplementary Figure S2 online). Collectively, our data showed that the richness and evenness of the microbiota were differentially reduced in the skin and oral cavity, but not in the gut, under AD conditions.

Closer lineage relationship between the skin and oral microbiota in AD patients

Next we characterized the similarities and differences in the composition of microbial communities between the samples (β-diversity) from AD patients and controls and among the different body sites. The microbial communities in the skin, oral cavity, and gut of both AD patients and healthy controls clustered separately (Figure 1c) and showed a pattern similar to that described in a previous report by the Human Microbiome Project Consortium (2012). However, the microbial community composition of AD patients was significantly different compared with that of healthy controls in all three body sites (permutational multivariate analysis of variance, $P < 0.001$) (Figure 1c, see Supplementary Table S2 online). The permutational multivariate analysis of variance results based on weighted UniFrac distances among these samples also corroborated these observations. The division of the samples into the three age groups mentioned yielded similar results (see Supplementary Table S2, Supplementary Figure S3 online).

We subsequently investigated the similarities and differences in the microbial communities between two body locations from the subjects with paired sequencing data (see Supplementary Table S1). The results showed that the

distance between the skin versus oral samples from AD patients was significantly reduced compared with that found for the controls, regardless of whether unweighted or weighted UniFrac distances were used (see Supplementary Figure S4 online; $P = 0.0236$ and $P = 4.673e\text{-}07$, Wilcoxon rank-sum test). The skin versus gut and oral versus gut distances were significantly reduced only if weighted and unweighted UniFrac distances were used, respectively (see Supplementary Figure S4; $P = 0.0346$, $P = 0.01561$, Wilcoxon rank-sum test). We further performed a SourceTracker analysis (Knights et al., 2011) and found that the resemblance between the skin and oral microbiota of AD patients was closer than that found for the controls, as demonstrated using oral samples as the "source" and skin samples as the "sink" (see Supplementary Figure S5 online).

Collectively, our data indicated a closer relationship in the microbial lineage between the skin and oral cavity of AD patients than between those of healthy controls.

Distinct abundance of taxa across the three body sites in AD

We found that the skin, oral cavity, and gut each harbored niche-specific taxa that were distinctly enriched in AD patients or healthy controls (see Supplementary Tables S3–S5 online). For example, at the phylum level, *Proteobacteria* was significantly enriched in the oral samples and significantly decreased in the skin of AD patients, whereas *Firmicutes* was significantly enriched in the skin of AD patients. (see Supplementary Table S3, Supplementary Figure S6 online). At the genus level, 116, 53, and 39 genera showed significant differences in the skin, oral cavity, and gut between AD patients and controls, respectively (adjusted $P < 0.05$) (see Supplementary Table S4, Supplementary Figure S7 online). A stratified analysis of samples from different age groups revealed 98%, 82%, and 74% overlap in the genera found in the skin, oral cavity, and gut among the different age groups, respectively. Considering the notable similarity in genera from each body site among the three age groups, the subsequent analysis were carried out mainly based on the combined age group. The top 15 genera with the most significant differences in abundance among the three habitats were shown in Figure 2, and 10, 8, and 8 genera were specific to the skin, oral cavity, and gut, respectively (Figure 2). The Random-Forest Classifiers that could select the most representative genera that discriminated AD patients from controls also showed that the genera selected from each habitat were distinctive (see Supplementary Figure S8 online). Specifically, 36 of the 53 oral-specific genera in AD exhibited either the opposite direction of enrichment to that observed in the skin or were not found at a significantly different abundance in the skin (see Supplementary Table S4). Thus, at the phylum and genus levels, specific microbes were enriched distinctly in the skin or oral cavity of AD patients.

To further identify the patterns of the differences in the microbial community between AD patients and controls across the three body sites, we established co-abundance associations among the genera from the three body sites and then clustered them into three co-abundance groups (see Supplementary Figure S9a, S9b online). The most dominant genera in the three groups were *Staphylococcus*, *Streptococcus*, and *Bacteroides* (see Supplementary Table S6 online). We then visualized the network of these genera among the three body sites in AD patients and controls according to the Spearman correlation coefficient (Figure 3). The results showed that each habitat had its own dominant cluster of microbes and revealed alterations in both the microbes and the associations among the three habitats in AD patients (Figure 3). Interestingly, 14 of the top 15 genera of the skin microbiota that showed significant differences in abundances, as shown in Figure 2, were classified as belonging to the *Staphylococcus*-dominated group, which was more prevalent in the skin. This feature was also found for the microbes of the oral cavity and gut.

Collectively, AD patients showed site-specific clustering and alterations in the abundances of microbes across the three major body habitats.

Association between genera and AD-related clinical parameters

The Shannon diversity of the skin microbiota was significantly inversely correlated with the Scoring Atopic Dermatitis (SCORAD) index (Figure 4, $P < 0.001$, Spearman's correlation). Interestingly, the Shannon diversity of the oral microbiota was significantly positively correlated with the SCORAD index, although the correlation was weak (Figure 4, $P = 0.011$, Spearman's correlation). The Shannon diversity of the gut microbiota did not exhibit a significant correlation with the SCORAD index (Figure 4, $P = 0.305$, Spearman's correlation), and no significant correlation between the total IgE level or the eosinophil count and the Shannon diversity was found at any of the three body sites (see Supplementary Figure S10 online).

At the genus level, the genera found in the skin, oral cavity, and gut were differentially correlated with AD-related clinical parameters, the SCORAD index, total IgE level, and eosinophil count (see Supplementary Figure S11 online). For example, *Staphylococcus* in the skin was positively correlated with the SCORAD index, in contrast to most of the other microbes in the skin; the microbes in the oral cavity were mostly positively correlated with the SCORAD index, and the direction of these correlations was opposite that of the correlations between most skin microbes and the SCORAD index; many of the microbes in the gut were significantly correlated with the total IgE level and eosinophil count, and *Phascolarctobacterium* and *Klebsiella* were the most significant genera. Different age group also had its specific correlation between specific genera and clinical parameters (see Supplementary Figure S12 online).

Inverse association in function prediction between the skin and oral microbiota in AD patients

Function profiles of the microbiota obtained for each sample were predicted through a phylogenetic investigation of the communities involving the reconstruction of unobserved states based on the closed-reference 16S ribosomal RNA gene sequence (Langille et al., 2013). Of the 296 phylogenetic investigation of communities by reconstruction of unobserved states–inferred Kyoto Encyclopedia of Genes and Genomes pathways, 108, 71, and 4 pathways showed significant differences between AD patients and controls in the skin, oral cavity, and gut, respectively (Figure 5, see Supplementary Table S7 online, linear discriminant analysis

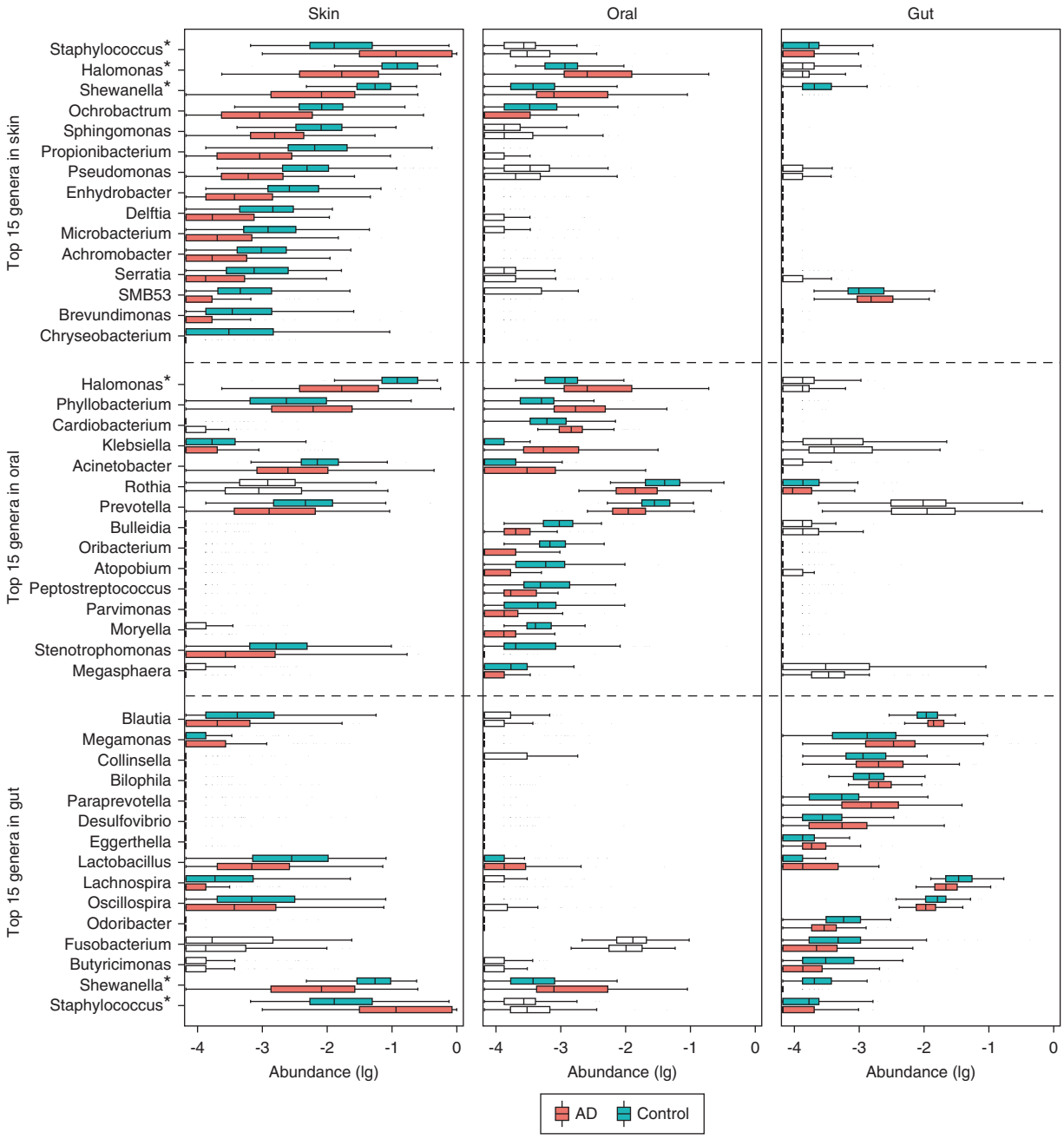


Figure 2. Differential abundance of genera in the three body sites. Relative abundances of the top 15 taxonomic genera that showed clear significant differences in abundances between AD patients and controls in the skin, oral cavity and gut. Only the significant genera are coloured. Adjusted $P < 0.05$, two-tailed Wilcoxon rank-sum test. *Denotes that the genus is ranked within the top 15 in at least two body sites. Skin (AD, n = 111; control, n = 102); oral (AD, n = 83; control, n = 77); and gut (AD, n = 88; control, n = 83). Boxplots represent the medians and interquartile ranges with whiskers, as determined by Tukey's method; black dots represent data points beyond the whiskers. AD, atopic dermatitis.

effect size, linear discriminant analysis score >2, $P < 0.05$). Interestingly, the enrichment direction of most of the pathways in the skin microbiota was opposite that found in the oral cavity (Figure 5). For example, the pathways for peptidoglycan biosynthesis, lysine biosynthesis, and galactose metabolism were enriched in the skin of AD patients and decreased in the oral cavity (Figure 5). In contrast, the pathway for tryptophan metabolism was enriched in the oral cavity of AD patients but decreased in the skin (Figure 5). Patterns of enrichment of the significantly different functional pathways of three age groups were similar to those of the combined age group (see Supplementary Figure S13 online). We also analyzed the correlation between the Kyoto Encyclopedia of Genes and Genomes pathways and AD-related clinical parameters (see Supplementary Figure S14 online), and significant correlations were found mainly between the pathways of the skin microbiota and the SCORAD index and eosinophil count (see Supplementary Figure S14). The total IgE level showed higher correlations with the pathways in the gut microbiota than with those in the skin and oral cavity. Most of the enrichment direction of the correlated pathways in the oral microbiota with respect to all three parameters

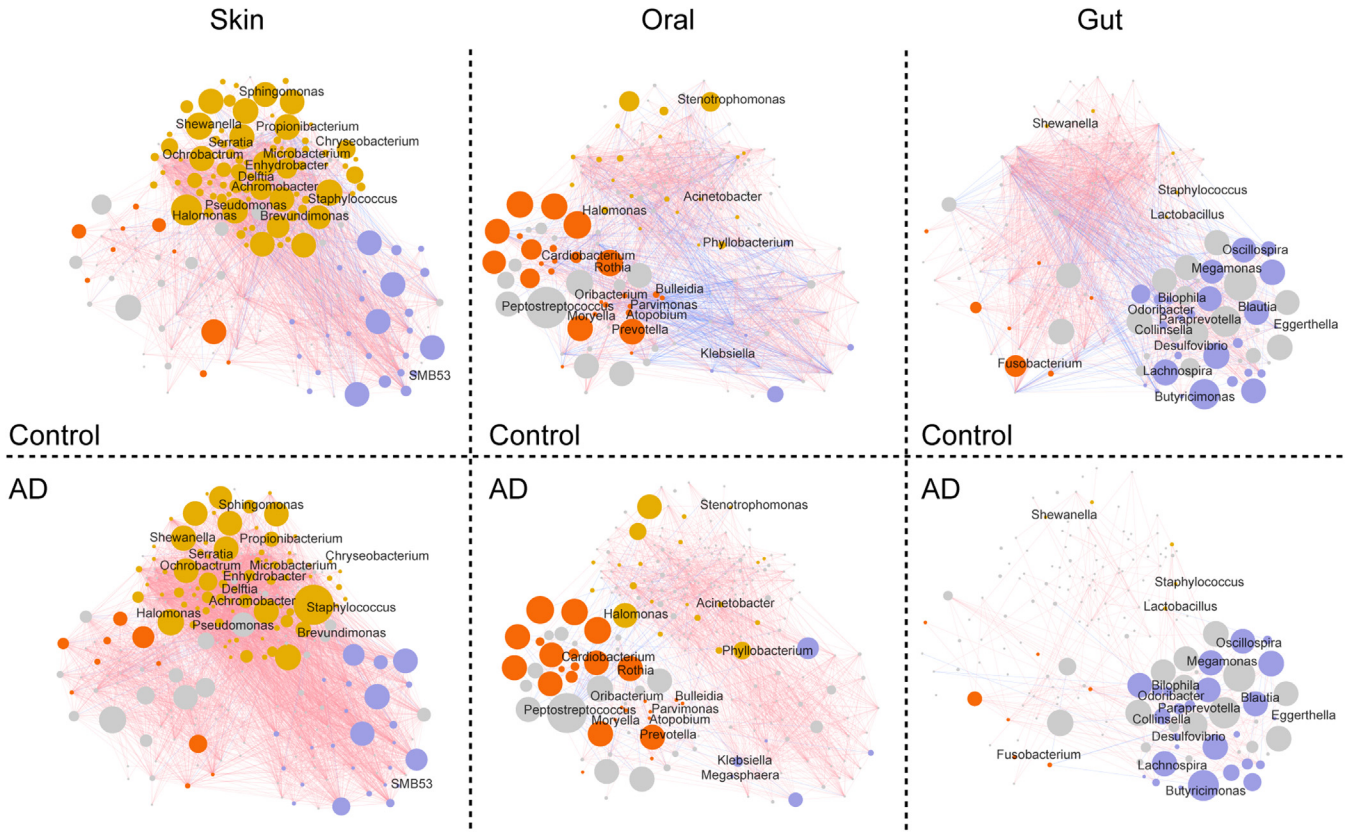


Figure 3. Co-abundance associations between genera in the three body sites. Network plots indicate the patterns in the variations of the genera between AD patients and controls across the three body sites. Each network shows a correlation between the genera in the respective data set (skin: AD, n = 111; control, n = 102; oral: AD, n = 83; control, n = 77; gut: AD, n = 88; control, n = 83). Each node represents a bacterial genus, and its dimension is proportional to the mean relative abundance across the three body sites. The three groups are named after the most abundant genera, and the color is coded as follows: *Staphylococcus*-dominated group (yellow), *Streptococcus*-dominated group (orange), and *Bacteroides*-dominated group (purple). The colors indicate that the genera are significantly different between AD patients and controls in the specific body site, and gray indicates no significant difference between AD patients and controls. Only the top 15 genera shown in Figure 2 are labeled. Blue lines, Spearman correlation coefficient < -0.3, adjusted $P < 0.05$; red lines, Spearman correlation coefficient > 0.3, adjusted $P < 0.05$. AD, atopic dermatitis.

was opposite that obtained for the skin microbiota (see Supplementary Figure S14).

DISCUSSION

In the present study, the microbial structure and function of the skin, oral cavity, and gut of AD patients were analyzed as an integrated community, which, to our knowledge, is previously unreported. The results showed that the taxonomic richness of the microbiota was differentially reduced in the skin and oral cavity of AD patients and revealed differential shifts in the composition of the microbial communities, particularly those of the skin and oral cavity, in AD patients; the different habitats in AD patients exhibited site-specific alterations in microbes, and an inverse association in the functional prediction was found between the skin and oral microbiota (Figure 6). The alterations of the microbiota in different body sites of AD patients were also differentially affected by age. Our data revealed the characteristics of the

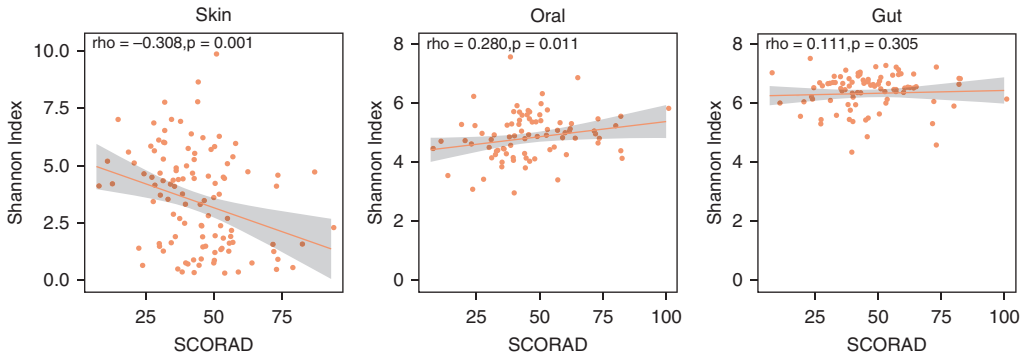


Figure 4. Associations of microbial diversity with SCORAD index. Scatter plot of Shannon Index against the SCORAD Index in the three body sites. The linear regression line and the 95% confidence interval are also visualized. The correlations between the Shannon Index and the SCORAD Index were calculated by Spearman's correlation analysis (skin AD, n = 111; oral AD, n = 83; gut AD, n = 88). AD, atopic dermatitis; SCORAD, Scoring of Atopic Dermatitis.

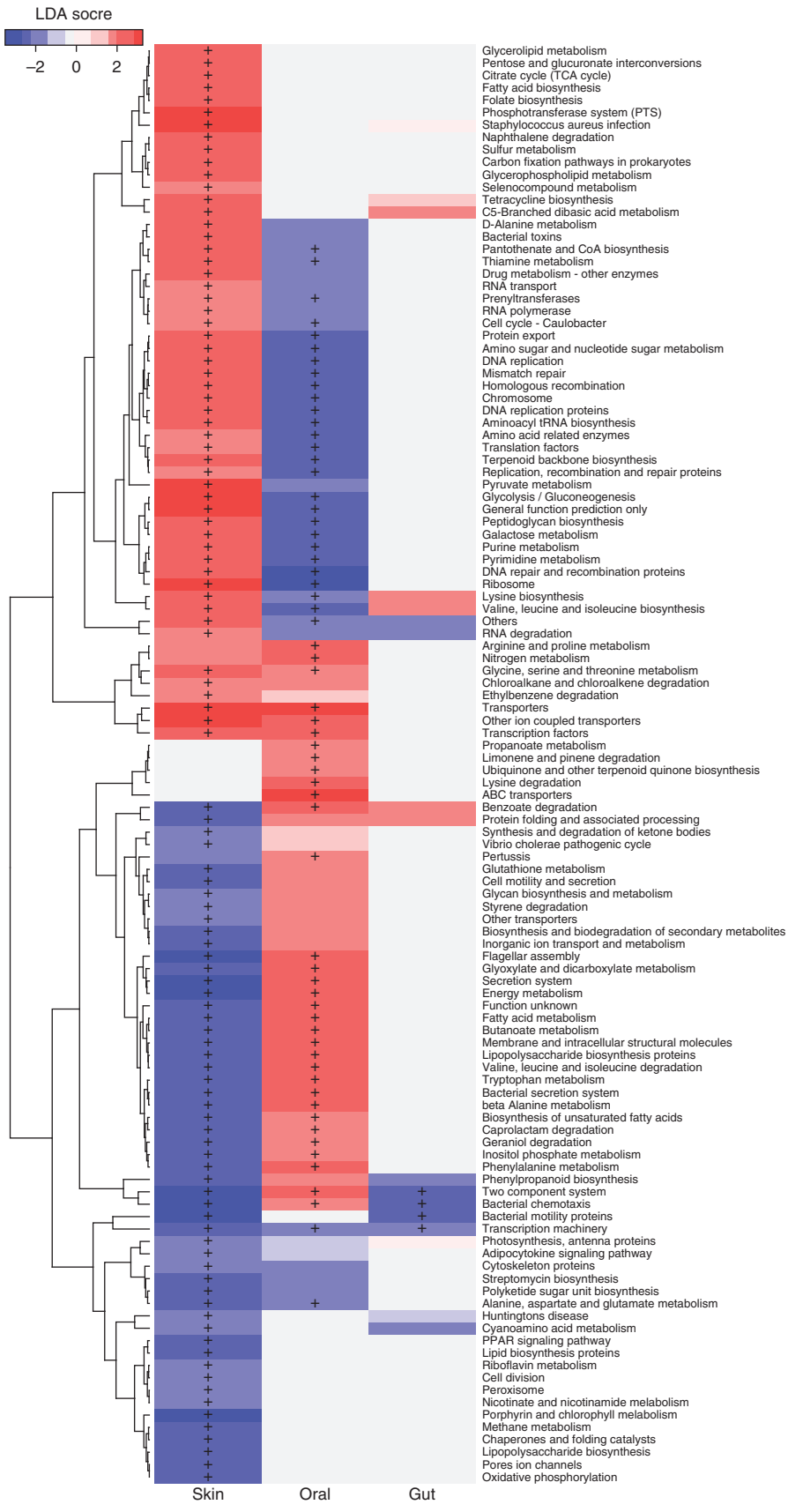


Figure 5. The inferred microbial functions in the three body sites. Heat map of the results of the LDA effect size performed on the pathway abundance between AD patients and controls across the three body sites (skin: AD, n = 111; controls, n = 102; oral: AD, n = 83; control n = 77; gut: AD, n = 88; control, n = 83). Blue, pathways enriched in the controls; red, pathways enriched in AD patients. +, linear discriminant analysis >2. AD, atopic dermatitis; LDA, linear discriminant analysis.

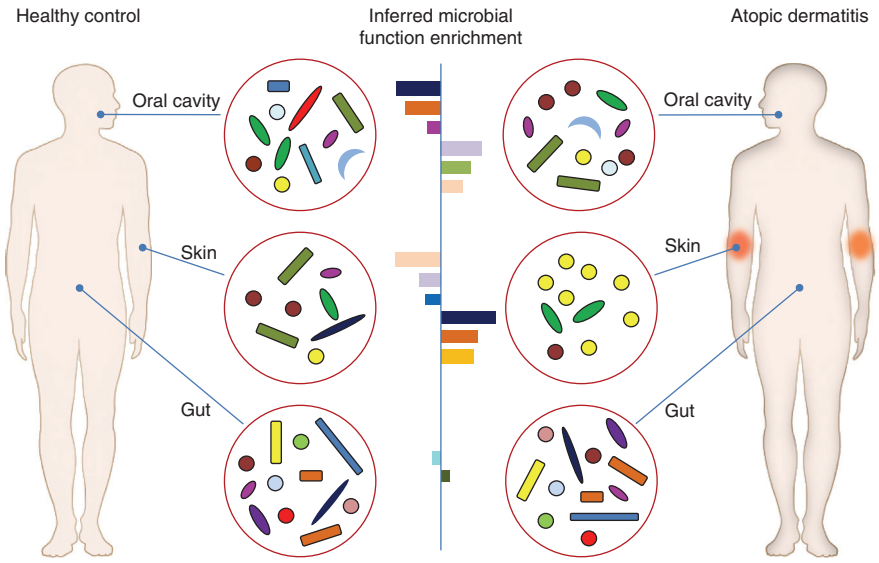


Figure 6. Summary for the structure and function of the microbiota in different body sites. There are differential reduction in the microbial diversity of the skin and oral cavity, but not the gut, of atopic dermatitis patients. The lineage distance between the skin and oral microbiota of atopic dermatitis patients is closer than that of the controls. Each habitat of atopic dermatitis patients has site-specific changes in microbes, and many of the microbes show opposing directions of enrichment in the skin and oral cavity. The inferred microbial function enrichment in the skin and oral cavity of atopic dermatitis patients is inversely associated.

oral microbiota in AD and enriched the understanding of the microbiota in different habitats of AD patients from a global point of view (Figure 6).

Our findings on the microbiota from the skin and gut of AD patients are generally consistent with those obtained in previous studies. The significant reduction in the skin microbial community diversity and the prominent abundance of *S. aureus* species in the lesions of AD patients are highly consistent with prior reports (Kong et al., 2012). The functional predictions based on the Kyoto Encyclopedia of Genes and Genomes pathways obtained in our study show high concordance with the metagenome data published by Chng et al. (2016). The unchanged α-diversity obtained in our study for the gut microbiota is similar to the results published by Song et al. (2016); however, many previous studies have observed a decreased microbial α-diversity in the gut (Abrahamsson et al., 2012; Arrieta et al., 2015; Fujimura et al., 2016). The reason for the discrepancy might be the age range of the enrolled patients, and both our study and the study conducted by Song et al. (2016) mainly included AD patients aged older than 1 year. Previous study also reported that alteration in specific microbes, such as *Clostridium difficile*, in the gut of infant might lead to eczema (Penders et al., 2007). Thus, our data indirectly confirm that the gut microbiota plays more important roles in the development of atopy during the early stages of life. However, the correlation between the genera in the gut microbiota and the increased IgE level obtained in our study (see Supplementary Figure S11) indicates that in adults, the gut microbiota might still be involved in the production of IgE.

The importance of the oral microbiota in diseases at distant sites has attracted increasing attention (Atarashi et al., 2017; Fan et al., 2018; Flemer et al., 2018; Flynn et al., 2016; He et al., 2015; Lucas et al., 2017; Yeoh et al., 2013). In the present study, we revealed a significant reduction in the microbial diversity of the oral cavity in AD patients and an apparent alteration in genera (Figures 1, 2). The alteration in the composition of the oral microbiota in AD in our study is different from that observed in oral diseases and chronic inflammatory diseases at distant sites, for example, inflammatory bowel disease. Dental caries show increased abundances of the genera *Prevotella* and *Atopobium*; in periodontal diseases, the abundances of the genera *Prevotella*, *Parvimonas*, and *Peptostreptococcus* are increased (He et al., 2015); in the oral cavity of inflammatory bowel disease patients, the level of *Prevotella* is increased (Lucas et al., 2017). These genera are hypothesized to produce a pro-inflammatory microenvironment in the oral cavity, which might result in local and systemic inflammation (He et al., 2015; Lucas et al., 2017). However, in the oral cavity of AD patients, the abundances of the genera mentioned were significantly decreased compared with those found in healthy individuals, which suggests that the oral microbiota in AD might play an anti-inflammatory role. Moreover, the direction of most of the Kyoto Encyclopedia of Genes and Genomes pathways associated with the oral microbiota was opposite that found for the skin microbiota (Figure 5), and this finding also suggests that the oral microbiota might have an anti-inflammatory function, which contrasts the pro-inflammatory functions of the *S. aureus*–dominant skin microbiota. Specifically, propanoate metabolism, butanoate metabolism, and tryptophan metabolism are enriched in the oral microbiota of AD patients (Figure 5). It has been reported that butyrate and propionate play an anti-inflammatory role by induction of regulatory T cells via inhibition of histone deacetylase (Arpaia et al., 2013; Sanford et al., 2016), and short-chain fatty acids also inhibit the activation of dendritic cells (Nastasi et al., 2015). We just reported that a metabolite of the tryptophan, IAId, attenuated the skin inflammation in a mouse model of MC903-induced AD-like dermatitis by inhibiting the expression of TSLP in keratinocytes in an AhR-dependent manner (Yu et al., 2018). The increased enrichment of these anti-inflammatory pathways in the oral microbiota of AD patients indicates that the oral microbiota of AD patients may play an anti-inflammatory role. Furthermore, sublingual immunotherapy is effective for the

treatment of atopic diseases (Halken et al., 2017), and it has been proposed that the mucosa of the oral cavity is found in a tolerogenic status (Novak et al., 2008). Thus, we speculate that the oral microbiota might create an anti-inflammatory microenvironment and play a regulatory role to counterbalance skin inflammation in AD.

A connection between the oral and gut microbiota has been proposed in gastrointestinal cancer and inflammatory bowel disease (Atarashi et al., 2017; Fan et al., 2018; Flemer et al., 2018), and multiple studies have shown that the oral cavity is the source of the microbiota of the nasal cavity (McLean, 2014) and lower respiratory tract (Segal et al., 2016). However, the relationship between the oral and skin microbiota under physiological and pathological conditions is unclear. Chng et al. (2016) showed that the majority of species enriched in AD-susceptible subjects are known commensals or opportunistic pathogens of the oral cavity. Our data demonstrate that the distance between the microbiota from the skin and oral cavity of AD patients is closer than that found in healthy controls (see Supplementary Figure S4), and a SourceTracker analysis demonstrated that the resemblance between the skin and oral microbiota of AD patients is higher than that found in healthy individuals (see Supplementary Figure S5). The inverse association of the functional pathways between the skin and oral microbiota also suggests a close relationship between the skin and oral microbiota in AD. Thus, we propose that the interplay between the oral and skin microbiota might be involved in the pathogenesis of AD.

Our study has several limitations. We only characterized the microbiota in different body sites at the time of AD flare up, and an analysis of the microbiota during the remission stage would give more information. The oral samples used in our study consisted of a mixture of the microbiota from the buccal mucosa, keratinized gingiva, supragingival plaque, and tongue dorsum, which includes the major sites of the oral cavity and provided abundant bacteria for the analysis; however, the current result might lack site specificity within the oral cavity. Age-associated alterations in the microbial structure and function in AD across the three body sites need further exploration, as the sample size for each age group in the present study is relatively small.

MATERIALS AND METHODS

Study subjects and clinical sample collection

A total of 172 AD patients were recruited, and 120 subjects without AD were selected from the set of volunteers (see Supplementary Table S1). None of the subjects had oral or skin diseases other than AD at the time of sampling, and none of the subjects had systemic diseases or received systemic treatment within 6 months before sample collection. Samples of the skin, oral cavity, and stool were collected from each patient and control. The subjects were instructed to avoid bathing and the application of topical medications and emollients at the sampling sites for 24 hours prior to sampling and to avoid brushing teeth and eating food for 2 hours prior to oral sampling. Skin samples were obtained from the lesions on both sides of the antecubital fossae from AD patients or from the same sites of healthy controls using a swab, as reported previously (Kong et al., 2012). Oral samples were collected by rubbing the buccal mucosa, keratinized gingiva, supragingival plaque, and tongue dorsum with a swab (Human Microbiome Project Consortium, 2012). Stool samples were self-collected, stored on ice, and sent to the laboratory within 1 hour of defecation. A diagnosis of AD was made based on the UK criteria (Guo et al., 2016), and the severity of AD was assessed using the SCORAD system. Peripheral blood was obtained for the preparation of serum and the analysis of blood cell counts. The total serum IgE levels were analyzed using the ImmunoCap System (Phadia, Uppsala, Sweden). The study was approved by the Ethics Committee of the Institute of Dermatology, Chinese Academy of Medical Science, and written informed consent was obtained from the participants or their parents.

Data availability statement

Sequence reads for the 558 samples have been deposited in the European Nucleotide Archive under the study number PRJEB23853. The authors declare that all other relevant data supporting the findings of this study are either available in this article and its Supplementary Materials online or from the corresponding author upon request.

Detailed information for DNA extraction and 16S ribosomal RNA amplicon sequencing, processing of 16S ribosomal RNA sequencing data, SourceTracker analysis, Random-Forest Classification, microbial structure network, prediction of functional potentials from 16S data, and statistical analysis are provided in Supplementary Material and Methods online.

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CONFLICT OF INTEREST

The authors state no conflict of interest.

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AUTHOR CONTRIBUTIONS

Conceptualization: XY (lead), WL (supporting), and CD (supporting); data curation: XX (lead), ZW (supporting), YX (supporting), XW (supporting); formal analysis: XX (lead), ZW (supporting), YX (supporting), XW (supporting); funding acquisition: XY (lead), WL (supporting) and HG (supporting); investigation: XY (lead), HW (supporting), CD (supporting), XL (supporting), YG (supporting), HS (supporting), JZ (supporting), and YH (supporting); methodology: XX (lead), ZW (supporting), YX (supporting), XW (supporting); project administration: XY (lead), WL (supporting) and HG (supporting); resources: XY (lead) and HG (supporting); supervision: XY (lead) and HG (supporting); validation: WL (lead) and XX (supporting); visualization: WL (lead) and XX (supporting); writing, original draft preparation: WL (lead) and XX (supporting); writing, review and editing: WL (lead), XY (supporting) and HG (supporting).

SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at https://doi.org/10.1016/j.jid.2019.02.009.

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SUPPLEMENTARY MATERIALS AND METHODS

DNA extraction and 16S ribosomal RNA amplicon sequencing

Total genome DNA from the skin, oral cavity, and stool samples was extracted by Novogene Bioinformatics Technology (Beijing, China) using the SDS method. The V3–V4 region of the 16S ribosomal RNA genes was amplified using the specific primer pair 341F/806R with barcoding. Sequencing libraries were generated using the TruSeq DNA PCR-Free Sample Preparation Kit (Illumina, San Diego, CA) following the manufacturer's recommendations, and index codes were added. Finally, the library was sequenced on an IlluminaHiSeq2500 platform, and 250-bp paired-end reads were generated.

Processing of 16S ribosomal RNA sequencing data

After the barcode and primer sequence were cut off, the chimera sequences were detected by comparing to the reference database (Gold database) using the UCHIME algorithm. The high-quality reads were then rarefied at 30,000 reads per sample to reduce the effects of variable sequencing depths on downstream analyses. Operational taxonomic units were generated with an identity cutoff of 97% using QIIME's UCLUST program, and the representative tags of each operational taxonomic unit were assigned taxonomy against the Greengenes database, version 13_8 (DeSantis et al., 2006) using the RDP classifier (Wang et al., 2007). The QIIME pipeline was also used to compute α- and β-diversity using the Shannon Index and weighted and unweighted UniFrac distances based on the operational taxonomic unit–level relative abundance.

To eliminate the influence of sequencing depth fluctuations, we downsized the number of reads to 30,000 for each sample. Fourteen samples (3 control skin samples, 1 AD oral cavity sample, and 10 AD skin samples) were excluded from the subsequent analysis due to an insufficient number of reads.

SourceTracker analysis to assess the potential seeding role of the microbiota between body sites

A SourceTracker analysis was performed to investigate whether AD inflammation could influence the communication of microbiota between different body sites (Knights et al., 2011). In each analysis, the samples from one of the three body sites were designated the sink, and the samples from the other two body sites were the selected source (Knights et al., 2011) (i.e., when the skin was the "sink," the oral cavity and gut were considered the "source"; when the oral cavity was the sink, the skin and gut were the source; and when the gut was the sink, the skin and oral cavity were the source).

Random-Forest Classification

A 10-fold cross-validation was performed using a Random-Forest Model (R 3.4.0, randomForest_4.6-10 package) with the genus abundance profiles of the AD and control samples. The cross-validation error curves from five trials of 10-fold cross-validation were averaged, and the minimum error in the averaged curve plus the standard deviation at that point was used as the cutoff of the classifier. All the genus markers selected by each cross-validation model were listed, and each genus was ranked according to its frequency of occurrence. The top ranking cutoff number of genera was then selected as the optimal value for the classifier (Zhang et al., 2015).

Microbial structure network

Associations between individual genera were evaluated using the Spearman correlation coefficient, visualized through hierarchical Ward clustering using Spearman correlation distance metrics and used to define co-abundant genera groups. Only significant associations (|coefficient |>0.3, adjusted *P*-value) were visualized in the background network. A Wiggum network plot for the three body sites in the AD cases and controls was created as described previously (Claesson et al., 2012) using Cytoscape software. The circle size represented the genus abundance, and the connections between nodes represent significant Spearman correlations between genera (|coefficient|>0.3, adjusted $P < 0.05$).

Prediction of functional potentials from 16S data

We used phylogenetic investigation of communities by reconstruction of unobserved states to generate a profile of putative Kyoto Encyclopedia of Genes and Genomes Orthology groups from the operational taxonomic unit data as described previously (Langille et al., 2013). The relative abundance of Kyoto Encyclopedia of Genes and Genomes pathways was obtained by summing the relative abundance of Kyoto Encyclopedia of Genes and Genomes Orthology groups belonging to these pathways. Linear discriminant analysis effect size, which combines a statistical test (Kruskal–Wallis test and pairwise Wilcoxon test) with linear discriminate analysis for feature selection and ranking features by effect size (Segata et al., 2011), was used to detect significant pathways. The threshold employed on the logarithmic linear discriminant analysis score for the discriminative pathway was 2.0.

Statistical analysis

Unless otherwise stated, statistical analyses were performed using R software. The differential diversity and abundance of phyla, genera, or species between AD patients and controls were tested using the two-tailed Wilcoxon rank-sum test. Only phyla, genera, or species that were present in at least 35% of samples from each body site were included in the test. The *P*-values were adjusted by the Benjamini-Hochberg correction for multiple tests when required. Permutational multivariate analysis of variance based on weighted and unweighted UniFrac distances was performed with 999 permutations in R (3.4.0, vegan package) to assess the impact of AD on the microbiota at the three body sites. The relationships between microbial data and clinical parameters were assessed using Spearman correlation test with the stat package in R.

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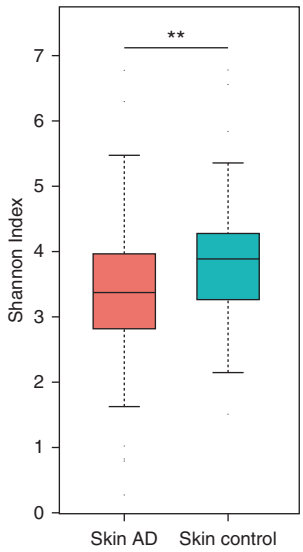
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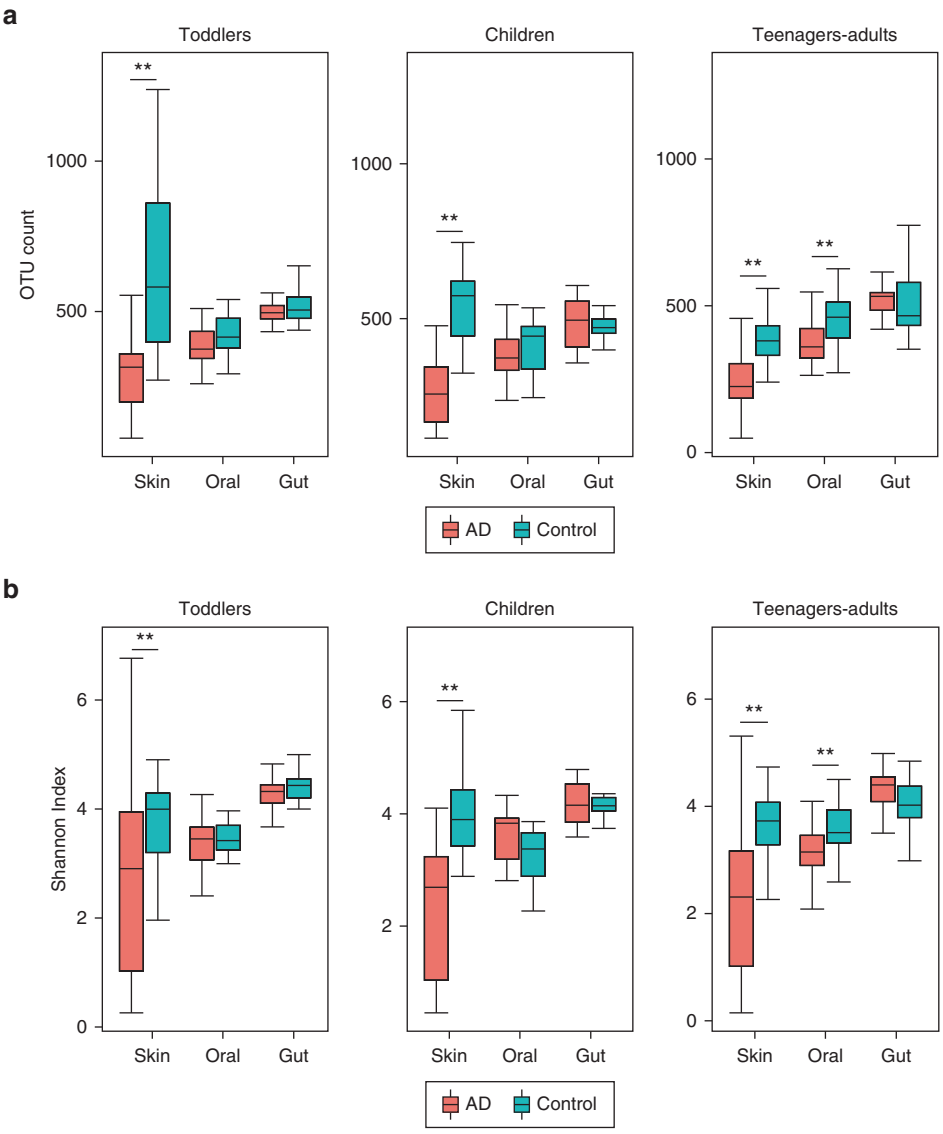
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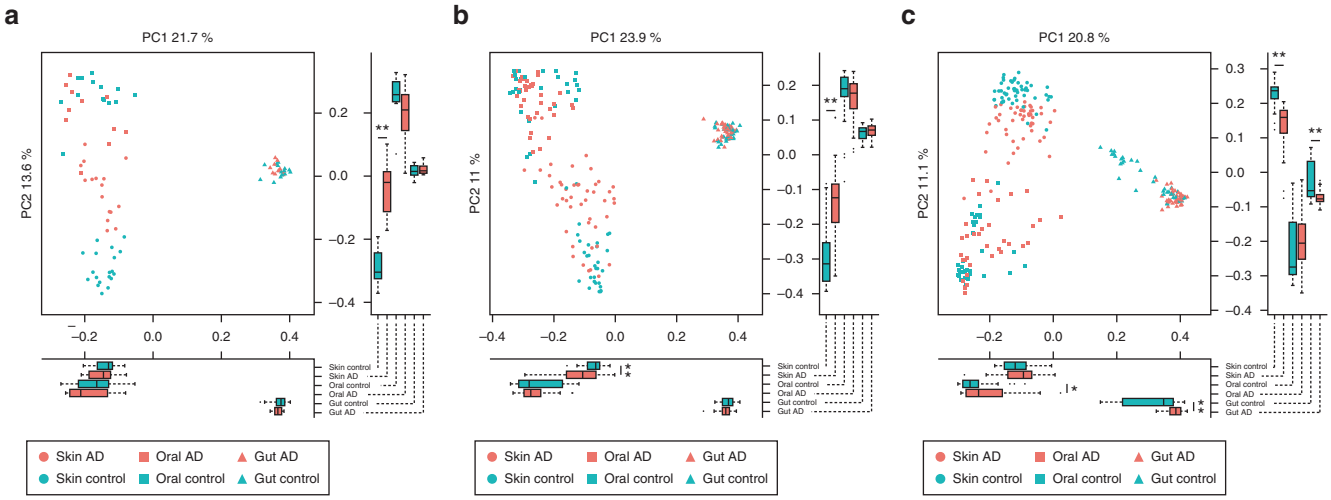
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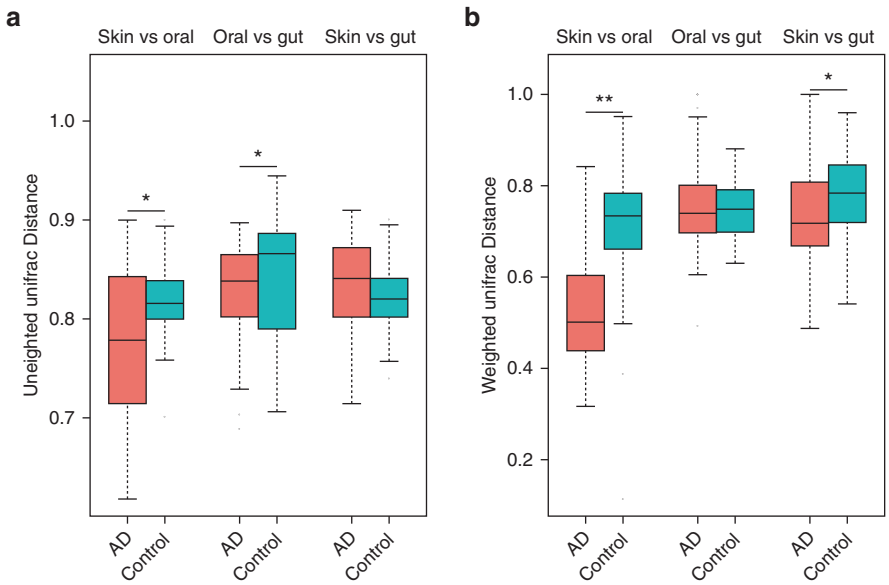
Supplementary Figure S1. α-Diversity of the skin microbiota with *Staphylococcus aureus* subtracted. The sequence of *S. aureus* was subtracted and α-diversity (Shannon Index) of the skin microbiota in AD patients and controls were analyzed. $^{**}P < 0.01$. Two-tailed Wilcoxon rank-sum test was used to determine significance. Boxplots represent the median and interquartile range with whiskers determined by Tukey's method; black dots represent data point beyond the whiskers. AD, atopic dermatitis.



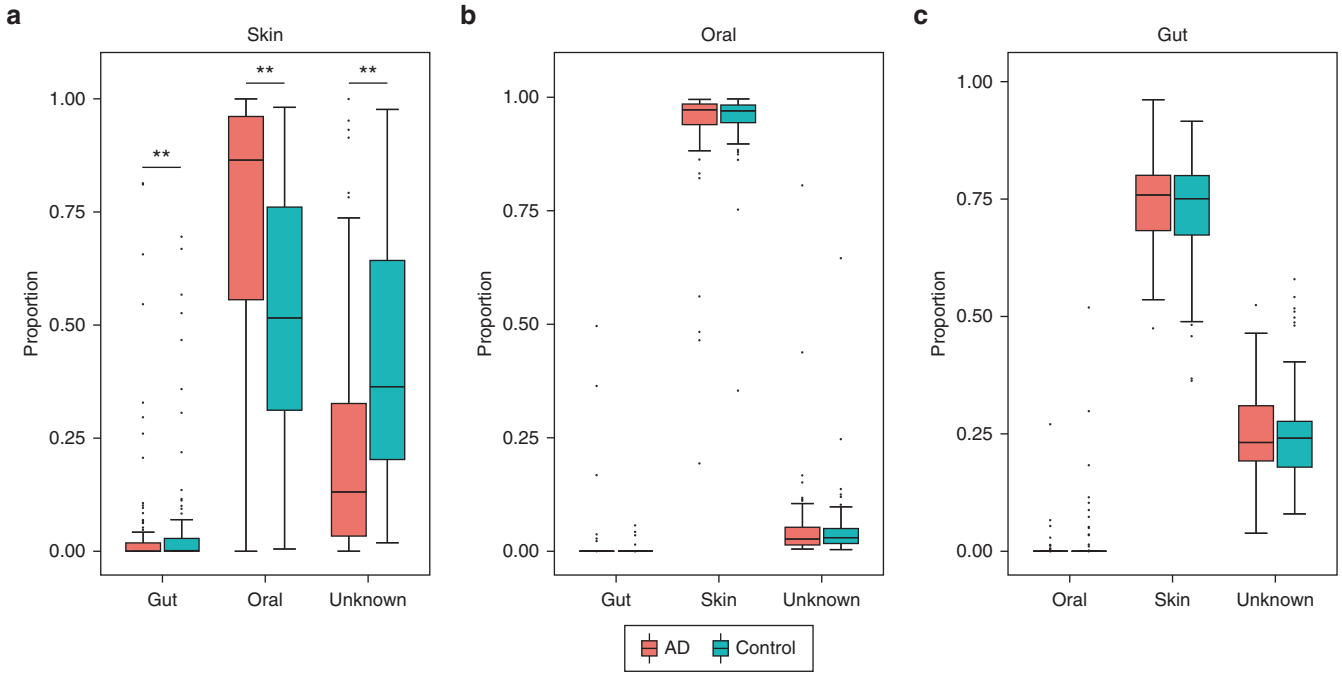
Supplementary Figure S2. Richness and diversity of the microbiota from the skin, oral cavity, and gut in three age groups. Richness (**a**) and α-diversity (Shannon Index) (**b**) of the microbiota from the three body sites at the OTU level in different age groups. Toddlers, 1−3 years: skin (AD, n = 16; control, n = 19), oral (AD, n = 11; control, n = 14), and gut (AD, n = 12; control n = 14); children, 4−12 years: skin (AD, n = 46; control, n = 30), oral (AD, n = 37; control, n = 28), and gut (AD, n = 37; control, n = 30); teenagers-adults, 13−59 years: skin (AD, n = 49; control, n = 53), oral (AD, n = 35; control, n = 35), and gut (AD, n = 39; control, n = 39). *Adjusted $P < 0.05$, **adjusted $P < 0.01$. Two-tailed Wilcoxon rank-sum test was used to determine significance. AD, atopic dermatitis; OTU, operational taxonomic unit.



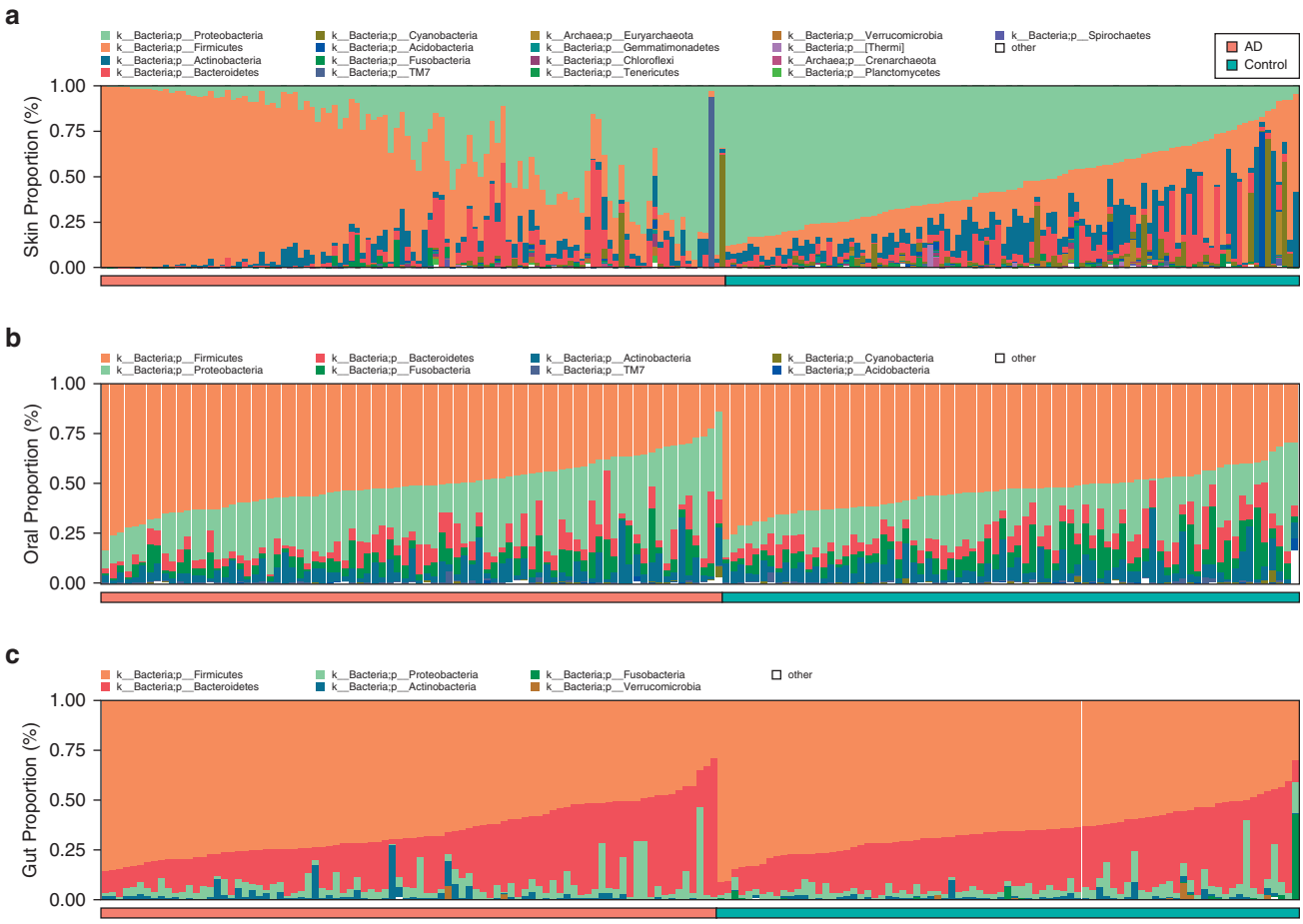
Supplementary Figure S3. PC analyses of the skin, oral cavity, and gut microbiota in three age groups. PC analyses on unweighted Unifrac distances is shown along the first two PC axes, stratified by age groups. (**a**) Toddlers, 1−3 years: skin (AD, n = 16; control, n = 19), oral (AD, n = 11; control, n = 14), and gut (AD, n = 12; control, n = 14). (**b**) Children, 4−12 years: skin (AD, n = 46; control, n = 30), oral (AD, n = 37; control, n = 28), and gut (AD, n = 37; control, n = 30). (**c**) Teenagers-adults, 13−59 years: skin (AD, n = 49; control, n = 53), oral (AD, n = 35; control, n = 35), and gut (AD, n = 39; control, n = 39). Each point represents a single sample. Box plots indicate the distribution of samples along the given axis. *$P < 0.05$, **$P < 0.01$. Two-tailed Wilcoxon rank-sum test was used to determine significance. AD, atopic dermatitis; PC, principal coordinate.



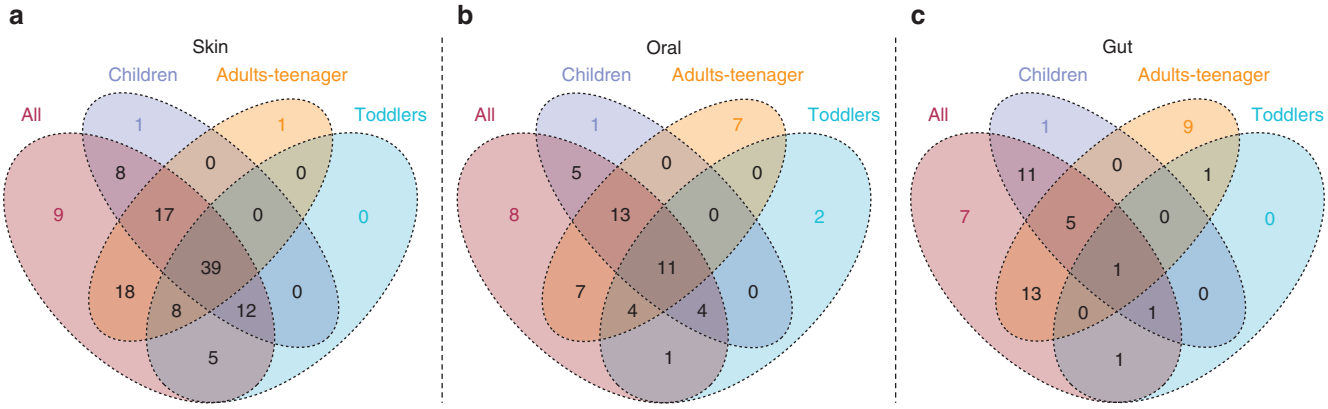
Supplementary Figure S4. Decreased site-specificity of the microbiota in AD patients. Boxplot of unweighted (**a**) and weighted (**b**) Unifrac distances between samples collected from the same individual (skin vs oral: AD, n = 34; control, n = 63; skin vs gut: AD, n = 36; control, n = 67; oral vs gut: AD, n = 78; control, n = 75). Data shown are distances between the gut and oral samples from the same individual, gut and skin samples from the same individual, and skin and oral samples from the same individual. *$P < 0.05$, **$P < 0.01$. Two-tailed Wilcoxon rank-sum test was used to determine significance. Boxplots represent the median and interquartile range with whiskers determined by Tukey's method; black dots represent data point beyond the whiskers. AD, atopic dermatitis.



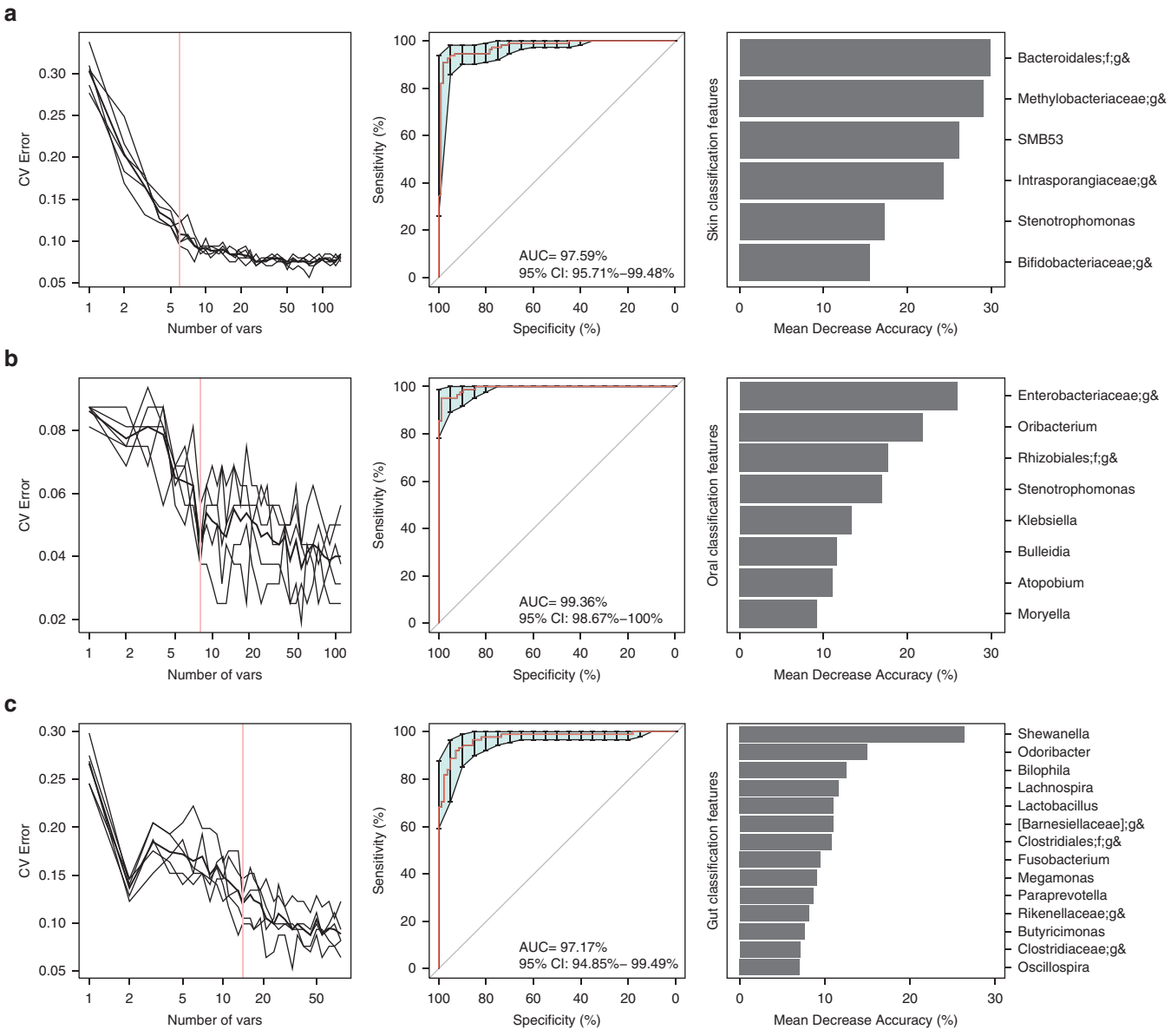
Supplementary Figure S5. Microbial taxa similarities between body sites in AD patients and controls. Boxplots of the proportion of operational taxonomic units in the skin, oral cavity, and gut were predicted to originate from the other two body sites in AD patients and controls, respectively. (**a**) The skin samples as the "sink" and the oral cavity or gut samples as the "source"; (**b**) the oral samples as the sink and the skin or gut samples as the source; (**c**) the gut samples as the sink and the skin or oral samples as the source. Skin: AD, n = 111; controls, n = 102; oral: AD, n = 83; control, n = 77; and gut: AD, n = 88; control, n = 83. $^{**}P < 0.01$. Two-tailed Wilcoxon rank-sum test was used to determine significance. Boxplots represent the median and interquartile range with whiskers determined by Tukey's method; black dots represent data point beyond the whiskers. AD, atopic dermatitis.



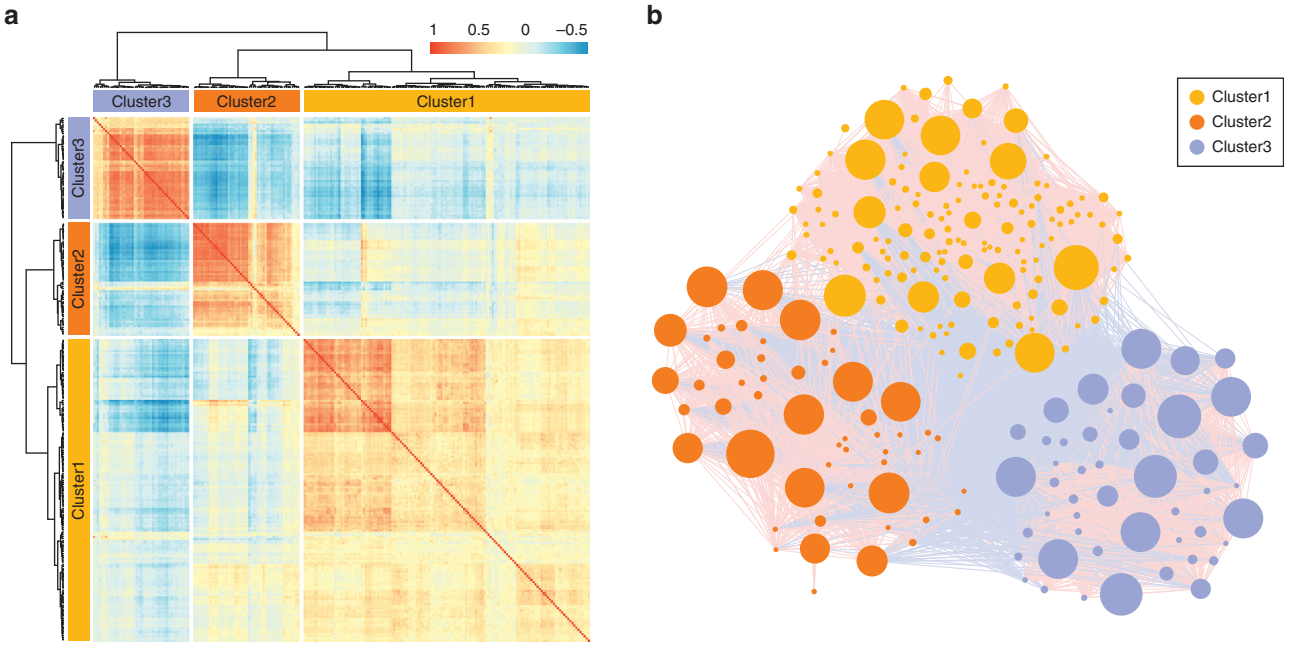
Supplementary Figure S6. Alterations of the microbiota in AD patients. Graph demonstrating differences of the phylum abundance between AD patients and controls in samples of the skin (**a**), oral cavity (**b**), and gut (**c**). Legend above the graph indicates the color for each phyla of bacterium represented.



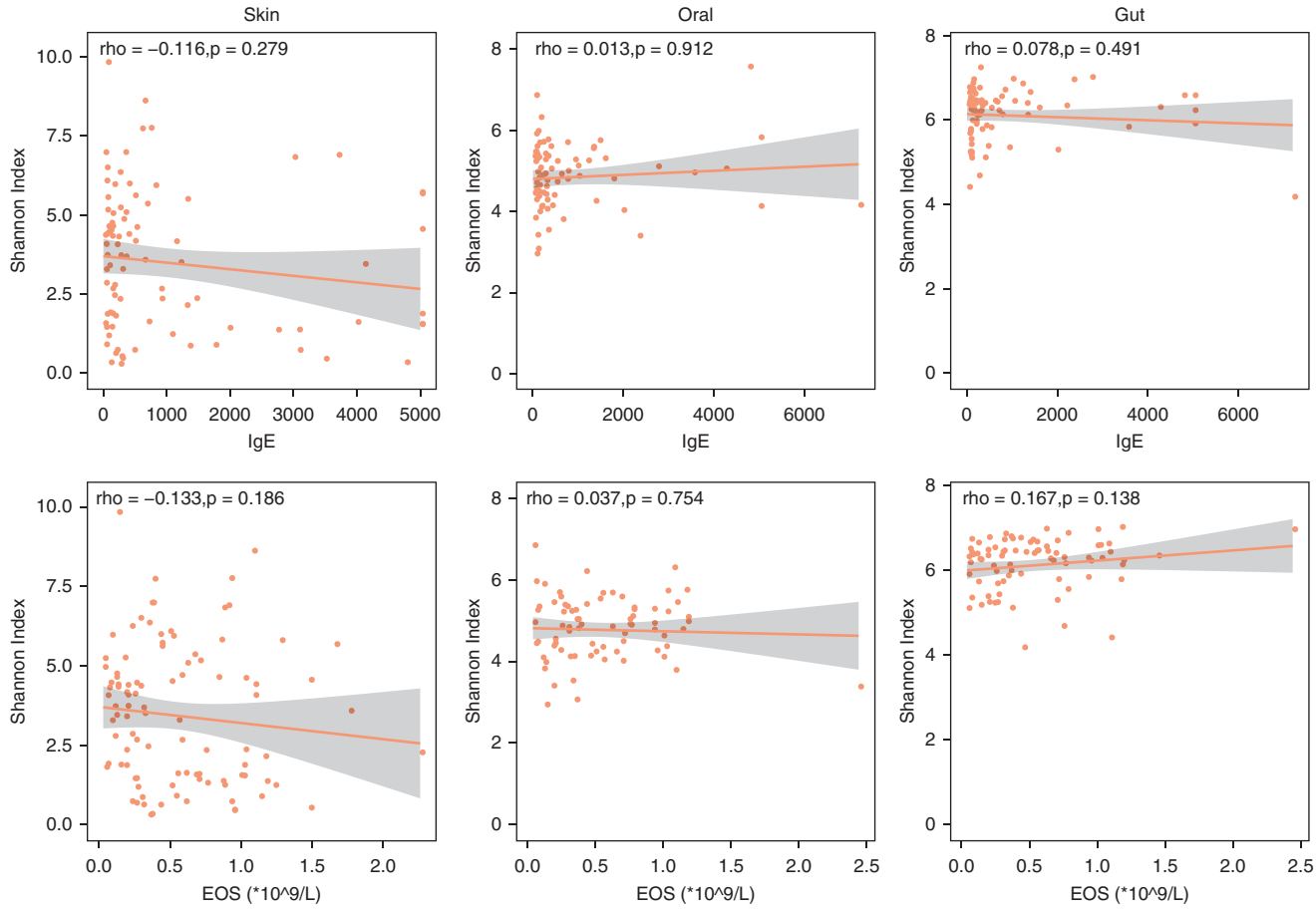
Supplementary Figure S7. Venn diagram of significantly different genera between AD patients and controls among different age groups. Venn diagram showing the unique and overlap signature genera among the different age groups in the skin (**a**), oral cavity (**b**), and gut (**c**), respectively. All, all samples: skin (AD, n = 111; control, n = 102), oral (AD, n = 83; control, n = 77), and gut (AD, n = 88; control, n = 83); Toddlers, 1–3 years: skin (AD, n = 16; control, n = 19), oral (AD, n = 11; control, n = 14), and gut (AD, n = 12; control, n = 14); Children, 4–12 years: skin (AD, n = 46; control, n = 30), oral (AD, n = 37; control, n = 28), and gut (AD, n = 37; control n = 30); teenagers-adults, 13–59 years: skin (AD, n = 49; control, n = 53), oral (AD, n = 35; control, n = 35), and gut (AD, n = 39; control, n = 39). Adjusted $P < 0.05$ was used to determine the significantly different genera. AD, atopic dermatitis.



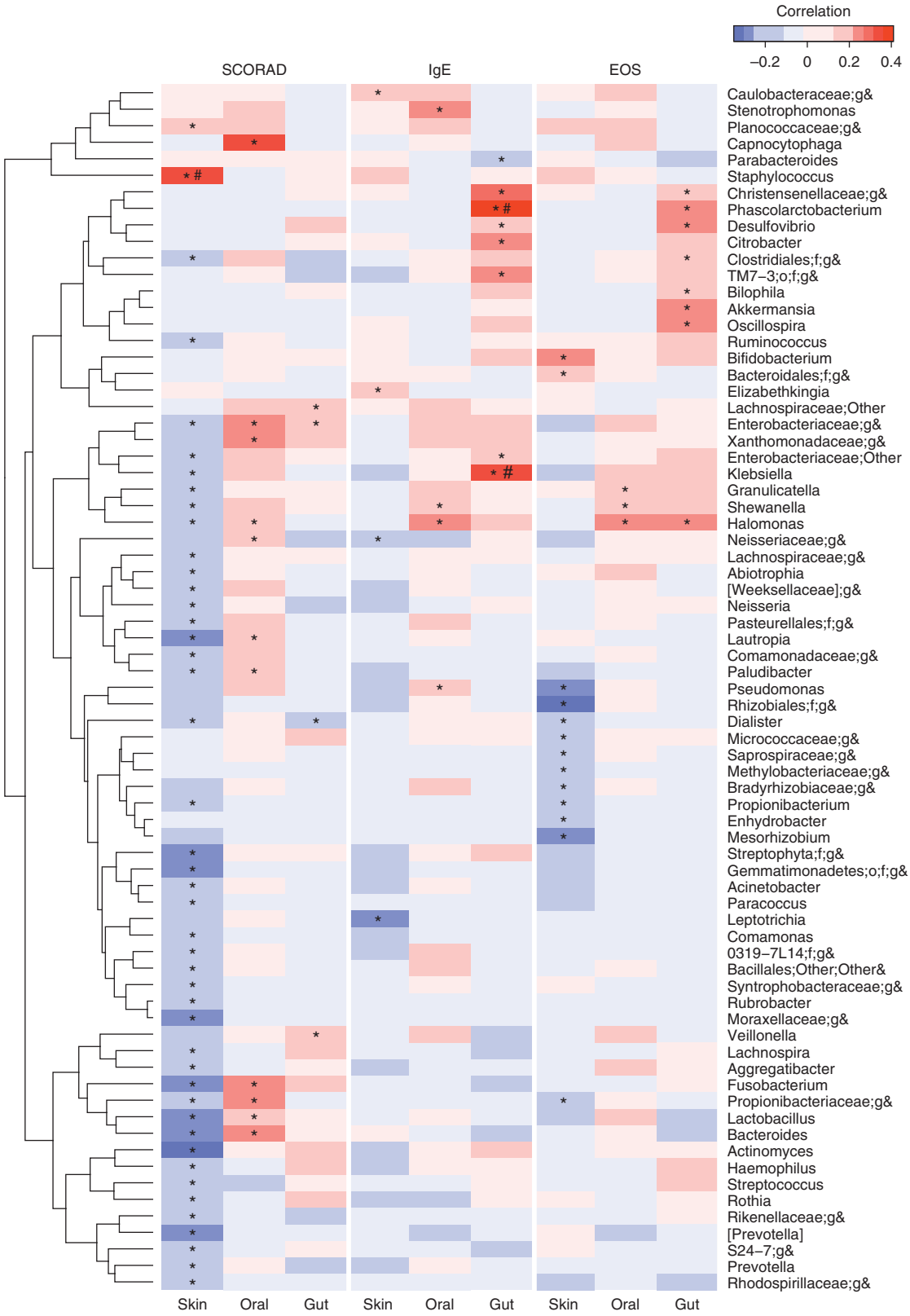
Supplementary Figure S8. Genus selected by the Random-Forest Classifiers. Random-Forest Classifiers was performed in three body sites (**a**, skin; **b**, oral; **c**, gut). Left in each penal, distribution of five trials of 10-fold CV error in Random-Forest Classification of atopic dermatitis patients versus controls samples as the number of genera increases. The gray curve indicates an average of the five trials (dark lines). The pink line marks the number of genera in the optimal set. Median in each panel, receiver operating curve for the result of the optimal set in three body sites. The 95% CIs are shown as shaded areas. Right in each panel, the optimal set of genera selected by the classifier, ranked by contribution to reduction of mean square error. skin: atopic dermatitis, n = 111; controls, n = 102; oral: atopic dermatitis, n = 83; control, n = 77; and gut: atopic dermatitis, n = 88; control n = 83. &, unclassified operational taxonomic units reported at higher taxonomic level; AUC, area under the curve; CI, confidence interval; CV, cross-validation.



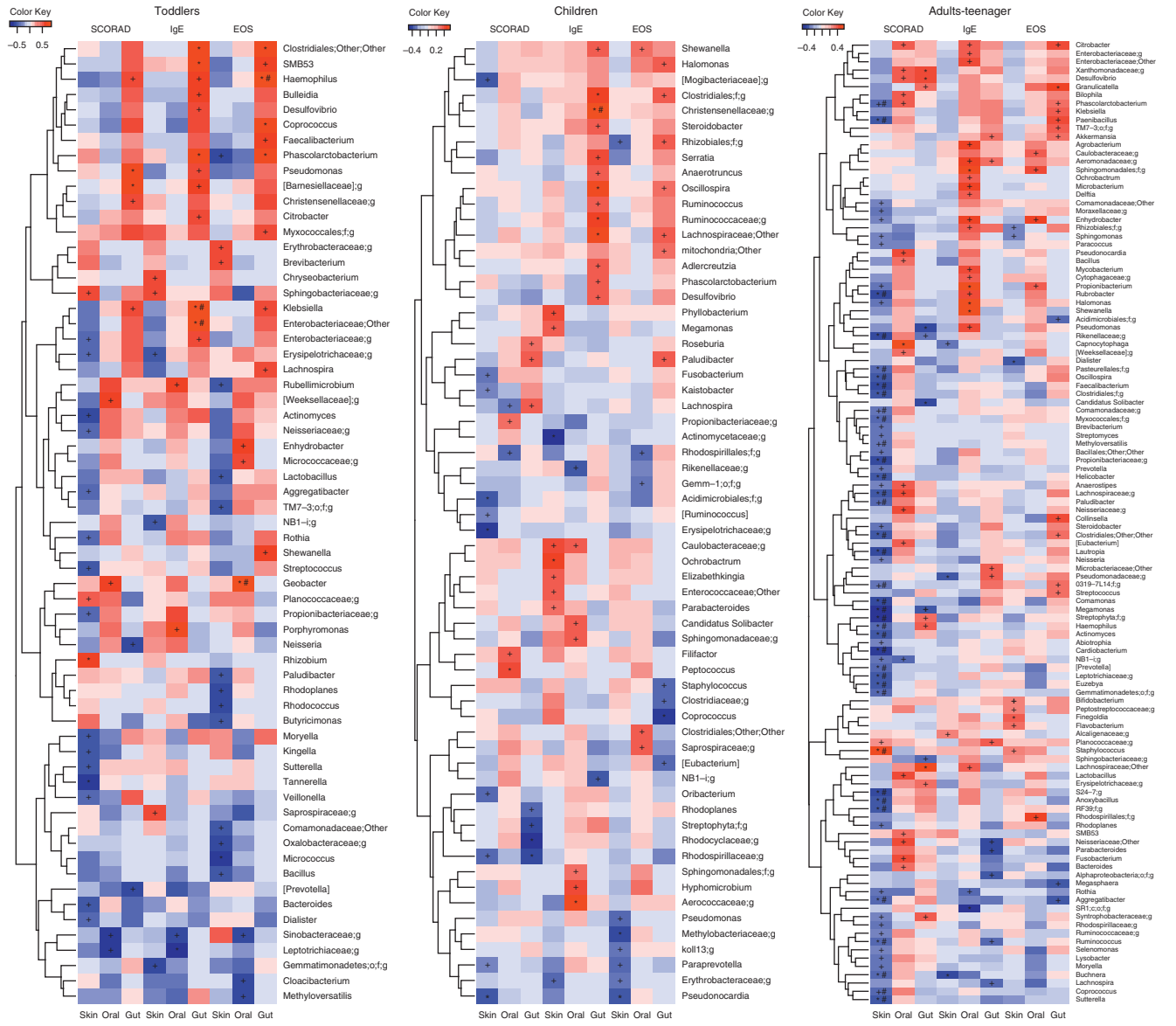
Supplementary Figure S9. Assignments of bacterial co-abundance groups. (**a**) Co-abundance groups assignment relied on a heat plot showing Spearman correlations between genera clustered by the Spearman correlation coefficient and Ward linkage hierarchical clustering. Colors are indicative of the three identified co-abundance groups. (**b**) Network plot correlations between the three co-abundance groups identified. Circle size is representative of the genus abundance and the connections between nodes represent significant spearman correlations between genera (|rho|>0.3, adjusted $P < 0.05$). Skin: AD, n = 111; controls, n = 102; oral cavity: AD, n = 83; control, n = 77; and gut: AD, n = 88, control, n = 83. Red line, positive correlation; blue line, negative correlation.



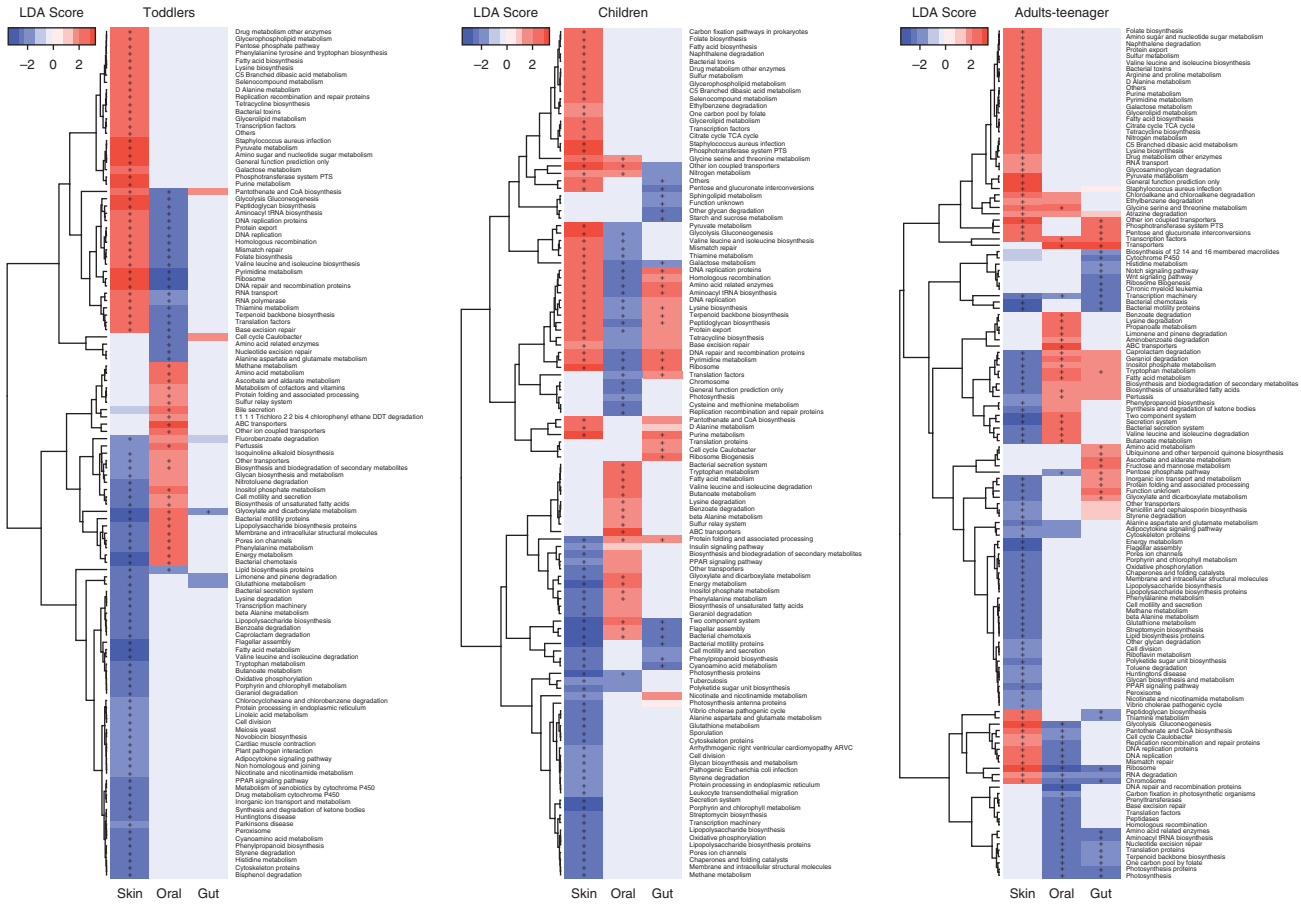
Supplementary Figure S10. Correlation between Shannon Index and clinical parameters. Scatter plot of Shannon Index against the IgE level or EOS number, stratified by body site. A linear regression line with 95% confidence interval is plotted to demonstrate the linear trend. EOS, eosinophil.



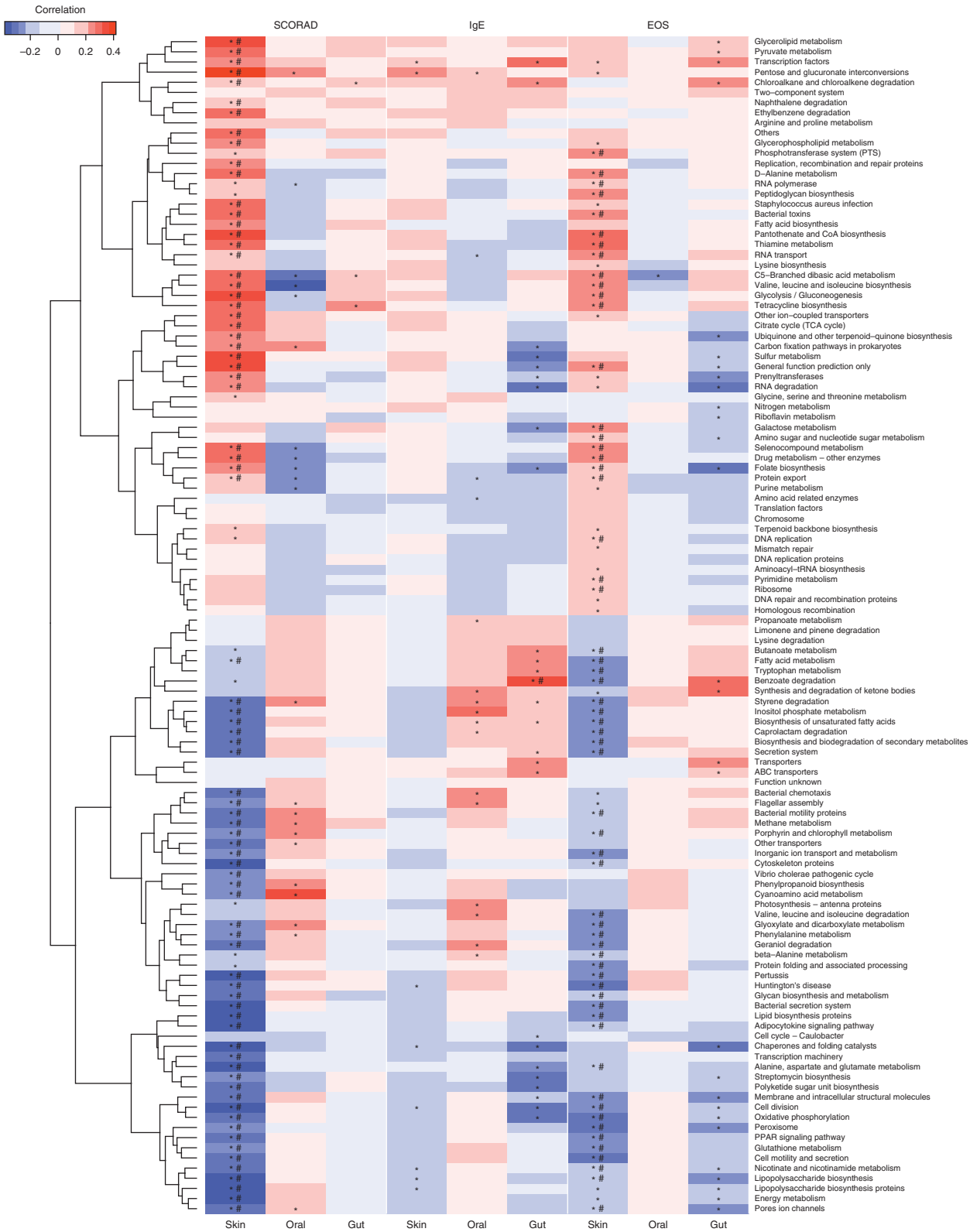
Supplementary Figure S11. Spearman correlation between the genera and the clinical parameters. Only genera correlated with at least one clinical parameter with $P < 0.05$ are shown. Skin: AD, n = 111; oral cavity: AD, n = 83; gut: AD, n = 88. *$P < 0.05$, #adjusted $P < 0.05$. &, unclassified operational taxonomic units reported at higher taxonomic level. AD, atopic dermatitis; EOS, eosinophil; SCORAD, Scoring of Atopic Dermatitis.



Supplementary Figure S12. Spearman correlation between the genera and the clinical parameters separated in three age groups. Only genera correlated with at least one clinical parameter with $P < 0.05$ are shown. Toddlers, 1−3 years: skin, AD, n = 16; oral, AD, n = 11; and gut, AD, n = 12. Children, 4−12 years: skin, AD, n = 46; oral, AD, n = 37; gut, AD, n = 37. Teenagers-adults, 13−59 years: skin, AD, n = 49; oral, AD, n = 35; gut, AD, n = 39. *$P < 0.05$, #adjusted $P < 0.05$. &, unclassified operational taxonomic units reported at higher taxonomic level. AD, atopic dermatitis; EOS, eosinophil; SCORAD, Scoring of Atopic Dermatitis.



Supplementary Figure S13. Shift in functions between atopic dermatitis patients and controls in three body sites of three age groups. Heat map of the results of LDA effect size performed on pathway abundance between atopic dermatitis patients and controls across the three body sites in different age group. Blue, pathways enriched in control; red, pathways enriched in atopic dermatitis. +, LDA >2. LDA, linear discriminant analysis.



Supplementary Figure S14. Spearman correlation between the inferred pathways and the clinical parameters. Only pathways correlated with at least one clinical parameter with $P < 0.05$ are shown. Skin: AD, n = 111; oral cavity, AD, n = 83; and gut: AD, n = 88. *P <0.05; #adjusted $P < 0.05$. EOS, eosinophil; LDA, linear discriminant analysis; SCORAD, Scoring of Atopic Dermatitis.